

Set / Weeks	Shanmukha — Robotics, Kinematics & Control	Ayush — Software, Algorithms & Documentation	Tanmay — Mechanical Design
SET 1 — Foundations W1–2	1. Finalize ROS 2 + Gazebo + MoveIt 2 setup 2. Set up robotics simulation environment 3. Define robot architecture and DOF configuration 4. Create decisions.md 5. Set initial kinematics/control requirements	1. Set up GitHub repository and branch structure 2. Set up project documentation structure 3. Literature survey — 20+ papers 4. Prepare literature comparison table 5. Identify research gap 6. Define evaluation metrics and baseline 7. Prepare initial project documentation/PPT material	1. Study UR5 flange and mounting requirements 2. Compare 3–4 candidate arms 3. Compare mass, reach, payload, DOF and URDF availability 4. Finalize selected arm
SET 2 — Bring-up & Interface Design W3–4	1. Bring up UR5 in Gazebo 2. Bring up selected small arm 3. Verify joint motion and basic planning 4. Study/implement manipulator kinematics for selected arms 5. Create placeholder coupling flange 6. Define robotics-side interfaces with Ayush	1. Create interfaces.md with Shanmukha 2. Define service/message structures 3. Implement mock reachability service 4. Create software architecture for coordination module 5. Set up GitHub development workflow 6. Start basic coordination code using mock data	1. Design UR5-to-dual-arm coupling flange 2. Parameterize arm spacing 3. Parameterize mounting tilt 4. Prepare preliminary mass/payload budget 5. Provide flange interface dimensions
SET 3 — Full Robot Assembly W5–6	1. Build UR5 + flange + 2 arms in Xacro 2. Configure robot TF structure 3. Configure MoveIt planning groups 4. Implement/verify manipulator kinematics 5. Configure ros2_control 6. Configure gripper control 7. Build CNC workcell 8. Add task frames 9. Achieve stable 16-DOF robot and joint control	1. Implement capability-map software 2. Implement workspace voxelization 3. Implement IK sampling/data storage pipeline 4. Implement manipulability calculation pipeline 5. Build capability-map visualization 6. Prepare software interface for real reachability service 7. Maintain GitHub integration and documentation	1. Finalize coupling-flange CAD 2. Export visual meshes 3. Create simplified collision meshes 4. Generate inertial properties 5. Build CNC workcell assets 6. Deliver simulation-ready CAD assets
SET 4 — Coordination & Baselines W7–8	1. Implement/verify robot forward & inverse kinematics pipeline 2. Implement real reachability service with robotics-side calculations 3. Implement task execution: target → plan → execute 4. Integrate coordination software 5. Build Condition A — single-arm baseline 6. Build Condition B — naive dual-arm 7. Verify complete CNC cycle	1. Replace mock service with real service 2. Implement carrier-placement solver 3. Implement greedy set-cover algorithm 4. Implement arm-assignment scoring 5. Implement hierarchical decision FSM 6. Integrate all coordination modules 7. Implement logging system 8. Ensure decisions/results are recorded	1. Integrate final mechanical geometry 2. Verify mounting clearances 3. Check arm-arm/workcell interference 4. Provide final dimensions, mass and CoG data 5. Support geometry corrections
SET 5 — Experiments & Evaluation W9–11	1. Complete robot-system integration 2. Debug kinematics/planning/control issues 3. Finalize CNC task sequence 4. Configure experimental robot states 5. Run trials with all three conditions 6. Perform mount-spacing/tilt simulation experiments 7. Investigate robotics-side failures	1. Implement experiment configuration scripts 2. Automate trial execution/data collection where possible 3. Run 25 randomized seeds 4. Process experimental logs using Python 5. Calculate cycle time, repositionings, joint travel, workspace utilisation and success rate 6. Generate plots 7. Perform statistical analysis 8. Perform failure classification 9. Maintain experiment dataset/results	1. Prepare exploded-view figure 2. Prepare CAD/render figures 3. Prepare mechanical configuration diagrams 4. Prepare workspace/mounting visualizations 5. Support experiment visualization
SET 6 — Documentation & Delivery W11–12	1. Finalize robotics methodology 2. Document kinematics, control and simulation architecture 3. Final integration testing 4. One-command demo launch 5. README technical content 6. Lead technical rehearsal	1. Complete software/algorithm documentation 2. Finalize plots and results 3. Maintain GitHub repository 4. Prepare PPT slides 5. Support final presentation design 6. Prepare demo interface/visualization if required 7. Technical report: coordination/software/analysis sections	1. Mechanical feasibility section 2. Final CAD figures 3. Final mechanical drawings 4. Produce demo video 5. Final visual polishing